

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Microcontrollers

STMicroelectronics bring sophisticated cyber protection to power-conscious connected devices with the new STM32L5 microcontroller (MCU) series featuring ARM Cortex-M33 core. Building on Cortex-M33, which boosts protection for small devices by integrating ARM's TrustZone hardware-based security, ST's STM32L5 series MCUs add further enhancements including flexible software isolation, secure boot, key storage and hardware cryptographic accelerators.

These also provide rich functionality, high performance and long run-times powered by coin cells or energy harvesting.

STMicroelectronics India
www.st.com

Buck boost regulator

RECOM presents RBB10 buck-boost converter with DOSA footprint and extra-low profile. This power module can reliably provide trimmed output voltages from a wide range of input voltages in applications such as USB voltage regenerators, 3.3V to 5V converters and battery regulators.

RBB10 series is a 4A non-isolated buck/boost regulator power module, which offers an extra-low profile for space-critical applications. Due to its buck-boost operation mode, it accepts input voltages that are higher, equal or lower than trimmed output voltage. Transition from buck to boost mode is smooth without any interruption to output. RBB10 module reaches an efficiency up to 96 per cent, which allows it to operate at full load in ambient temperatures as high as 85°C without forced air cooling.

RECOM Power

www.recom-power.com

DAC ICs

ROHM has launched a branding MUS-IC reflecting its focus on solutions for high-fidelity audio products. This branding covers a range of DA converter ICs (DAC ICs), optimised for high-fidelity reproduction and power supply products. These DAC chips, which deliver class-leading low noise and low distortion (SNR and THD + N characteristics) as demanded by audio equipment manufacturers, have been optimised for sound quality through repeated listening tests and laboratory measurements.

ROHM Semiconductor India Pvt Ltd
www.rohm.com

Transducers

KUSAM-MECO transducers are used for accurate measurement of electrical parameters, remote monitoring and automation, suitable for sinusoidal and true-RMS inputs, with high accuracy (0.2 per cent). These have high dielectric strength of 2.6kV AC minimum, between input/output/power case, and separate or inbuilt (self-powered) AC or DC auxiliary supply.

The transducers are suitable for single-phase as well as three-phase balanced or unbalanced load, in addition to precision measurement even for distorted wave. Various types of transducers are AC current, AC voltage, watt or var, combined watt and var, watt hour or var hour, combined watt and watt hour or var and var hour, and power factor.

Kusam Electrical Industries Ltd

www.kusamelectrical.com, kusam-meco.co.in

Digital signal controller

System designers looking for digital signal processing power with the design simplicity of an MCU can now utilise the 16-bit digital signal controller from Microchip Technology. Designed to deliver

faster deterministic performance in time-critical control applications, dsPIC33CK DSC has expanded context-selected registers to reduce interrupt latency and new, faster instruction execution to accelerate digital signal processor routines. The dsPIC33CK single-core family complements the recently-announced dsPIC33CH dual-core family based on the same core.

Microchip Technology Inc.
www.microchip.com

Digital power modules

Renesas Electronics has announced a family of encapsulated digital DC-to-DC PMBus power modules. The five RAA210xxx simple digital power modules offer advanced digital telemetry and performance, and are as easy to use as Renesas' analogue power modules. These are complete step-down regulated power supplies that deliver 25A, 33A, dual 25A, 50A and 70A of output current, while operating from 12V or 5V input power rails.

The RAA210xxx family provides point-of-load conversion for advanced FPGAs, DSPs, ASICs and memory used in servers, storage, optical networking and telecom equipment. Each device integrates a PWM controller, MOSFETs, inductor and passives inside a thermally-optimised, high density array encapsulated module.

Renesas Electronics Corp.
www.renesas.com

Network fuse

Littelfuse has introduced 40mA PICO 242 series hazardous area barrier network fuse, which supports designing barrier circuits for intrinsic safety products to prevent fires and explosions in dangerous atmospheres. The

